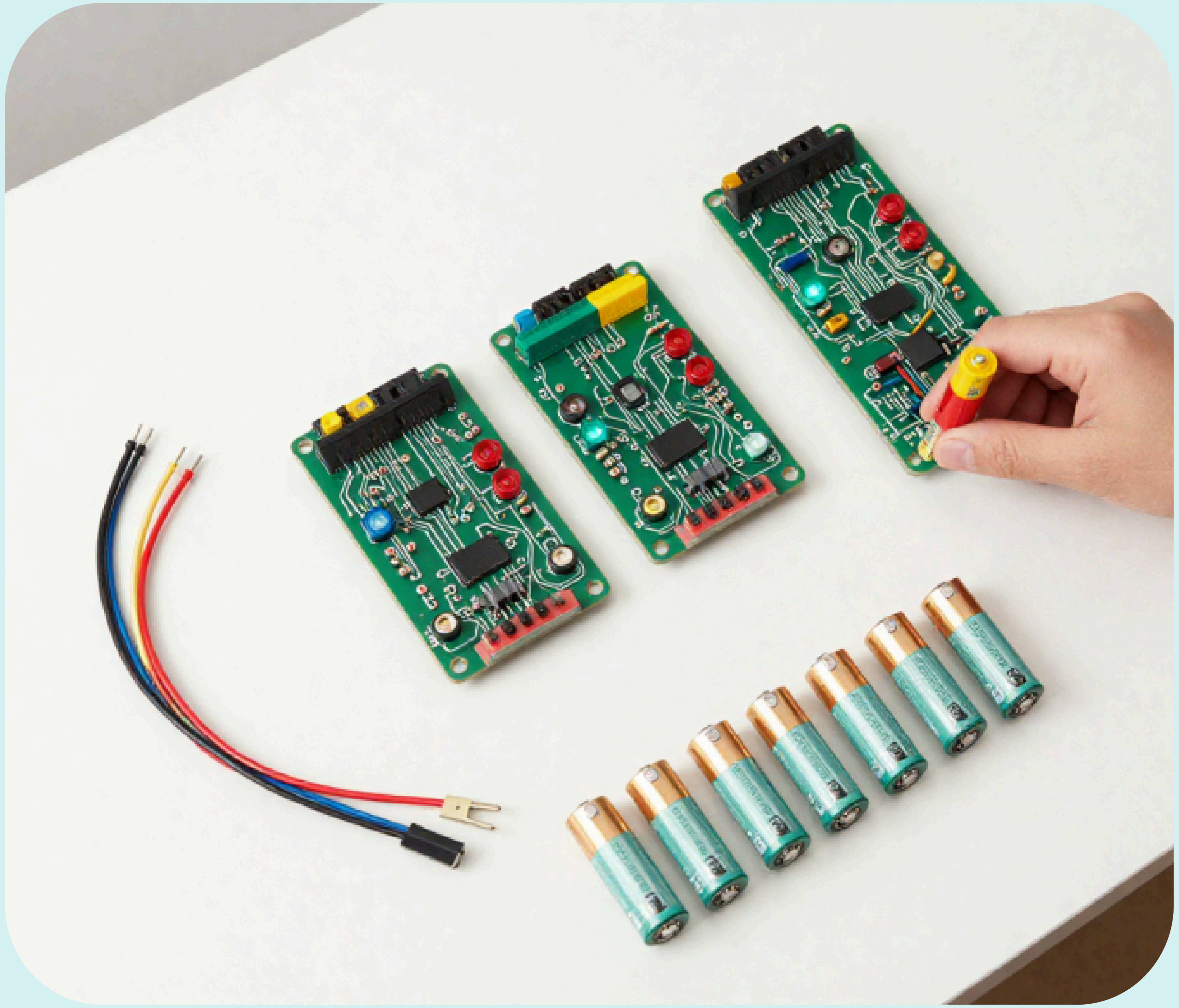
Breadboard Basics

Build Your First LED Circuit

What Is Electronics?

Electronics is all about tiny parts that help us make cool things like lights, sounds, and even robots!

These parts use electricity to work. When we connect them the right way, amazing things happen!

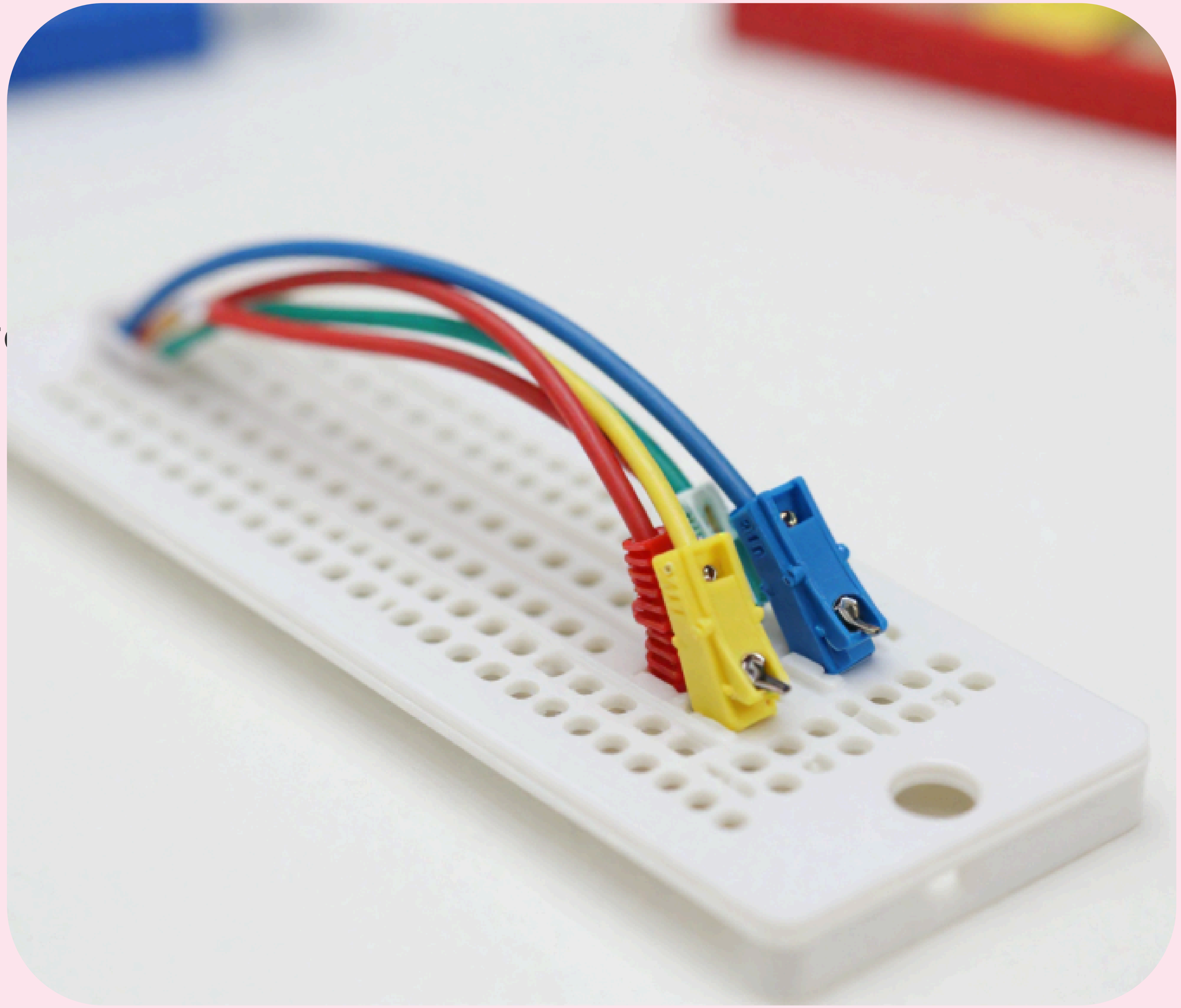


Meet the Breadboard

A breadboard is a special board that helps us connect electronic parts without soldering.

Just plug parts in and make circuits easily!

It has rows of tiny holes where you push in wir and components.



Components You Need

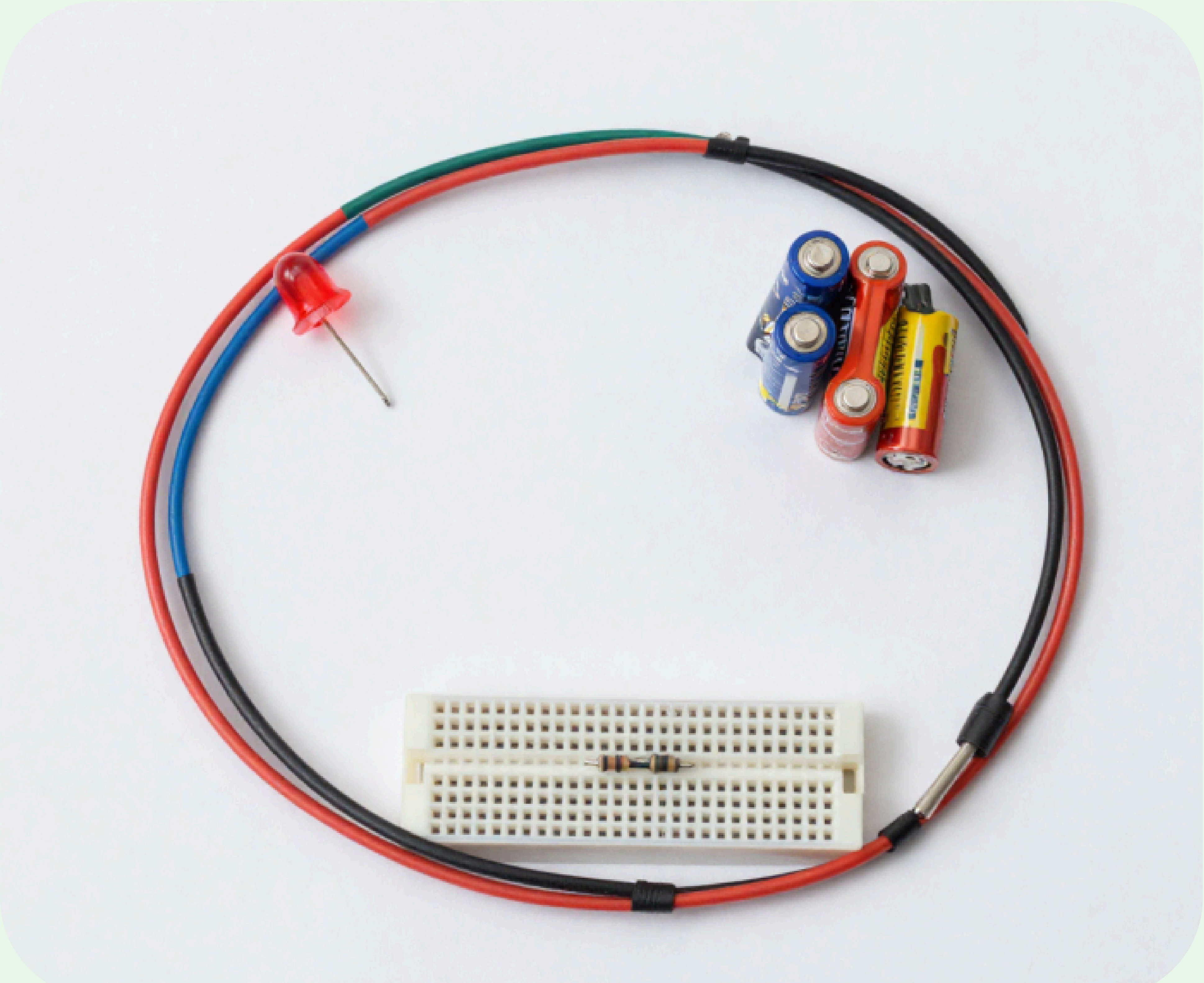
LED - lights up when electricity flows

Resistor - protects the LED from burning

Battery - gives power to the circuit

Jumper wires - connect parts together

Breadboard - holds everything in place



Safety First!

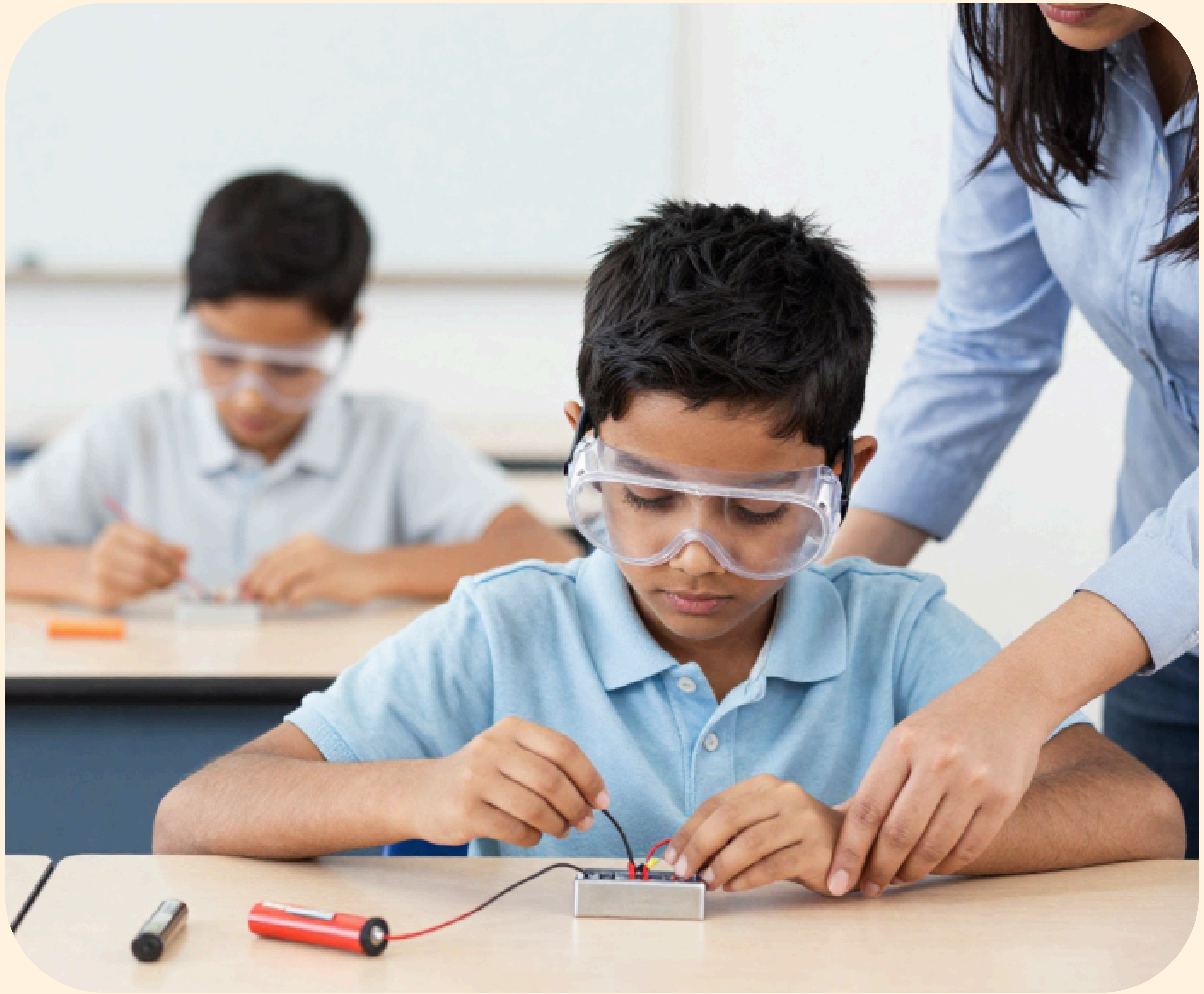
Never touch wires when the battery is connected

Always ask an adult for help

Keep components dry and clean

Handle the battery carefully

Never put small parts in your mouth

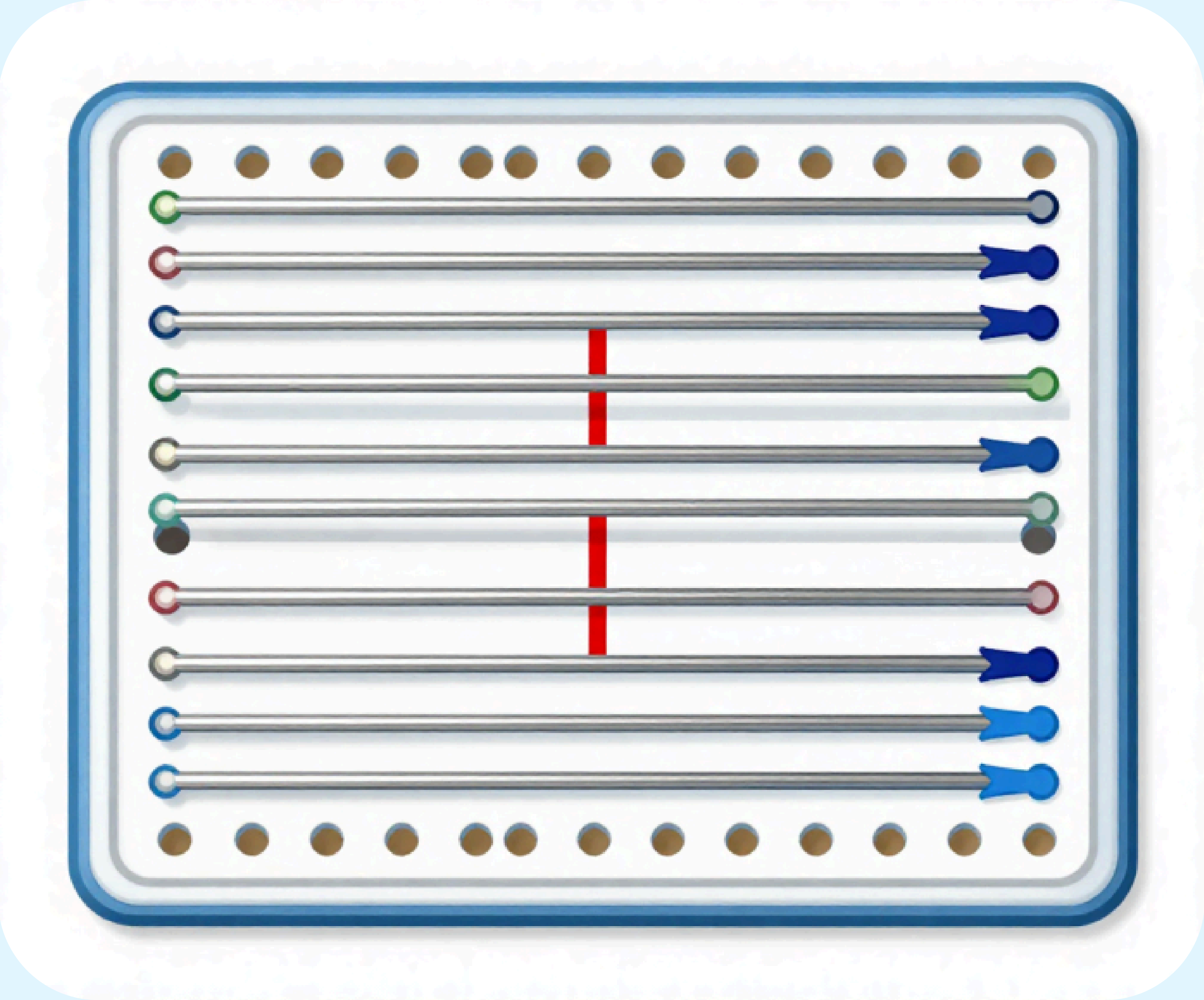


How the Breadboard Works

Inside the breadboard, tiny metal strips connect holes in rows and columns.

Parts plugged in the same row are joined together!

The two long strips on the sides are for power (+) and ground (-).

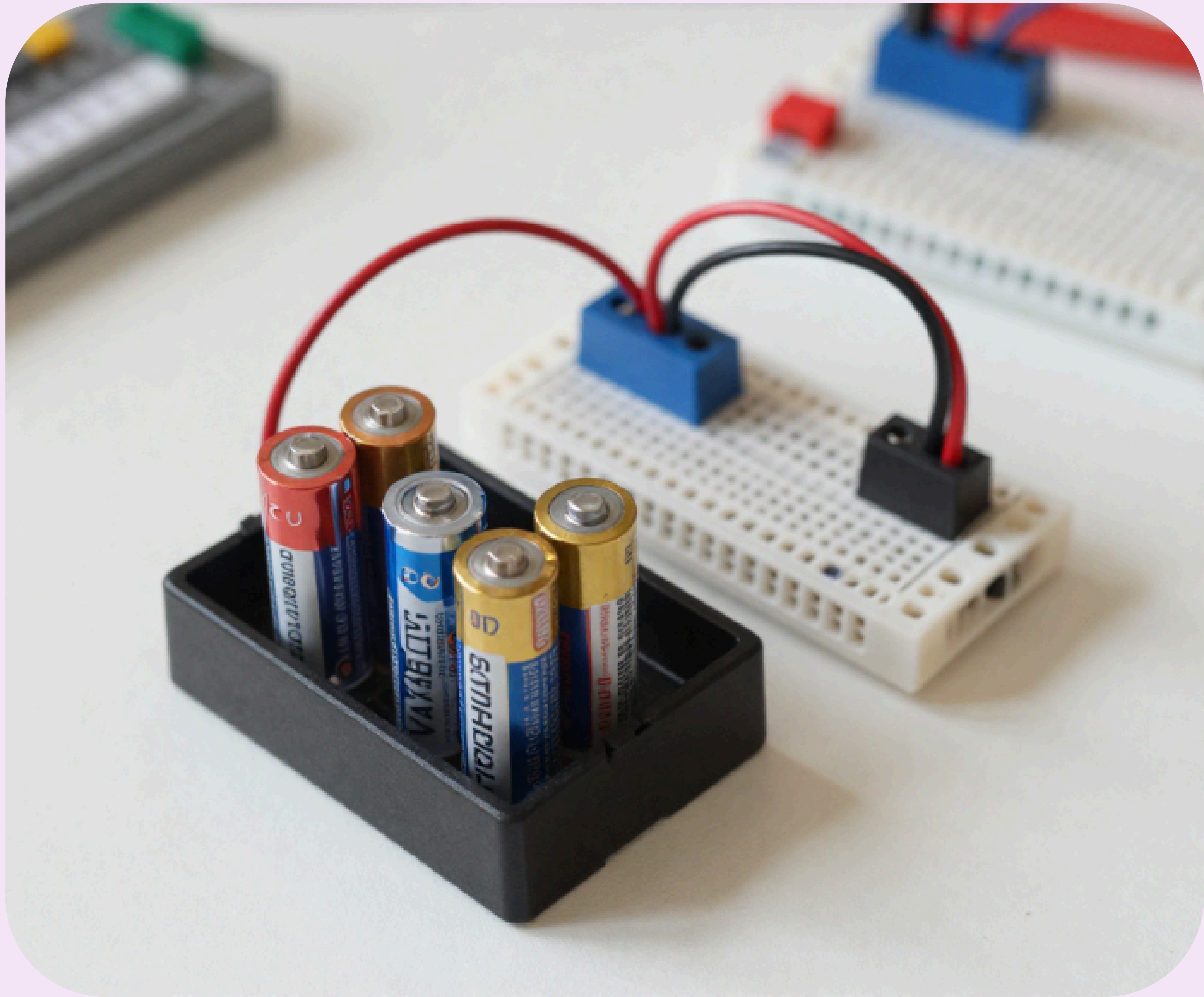


Step 1 — Place the Battery

Put the battery holder next to the breadboard to give power.

Find the positive (+) and negative (–) ends.

The red wire goes to + and the black wire goes to –.

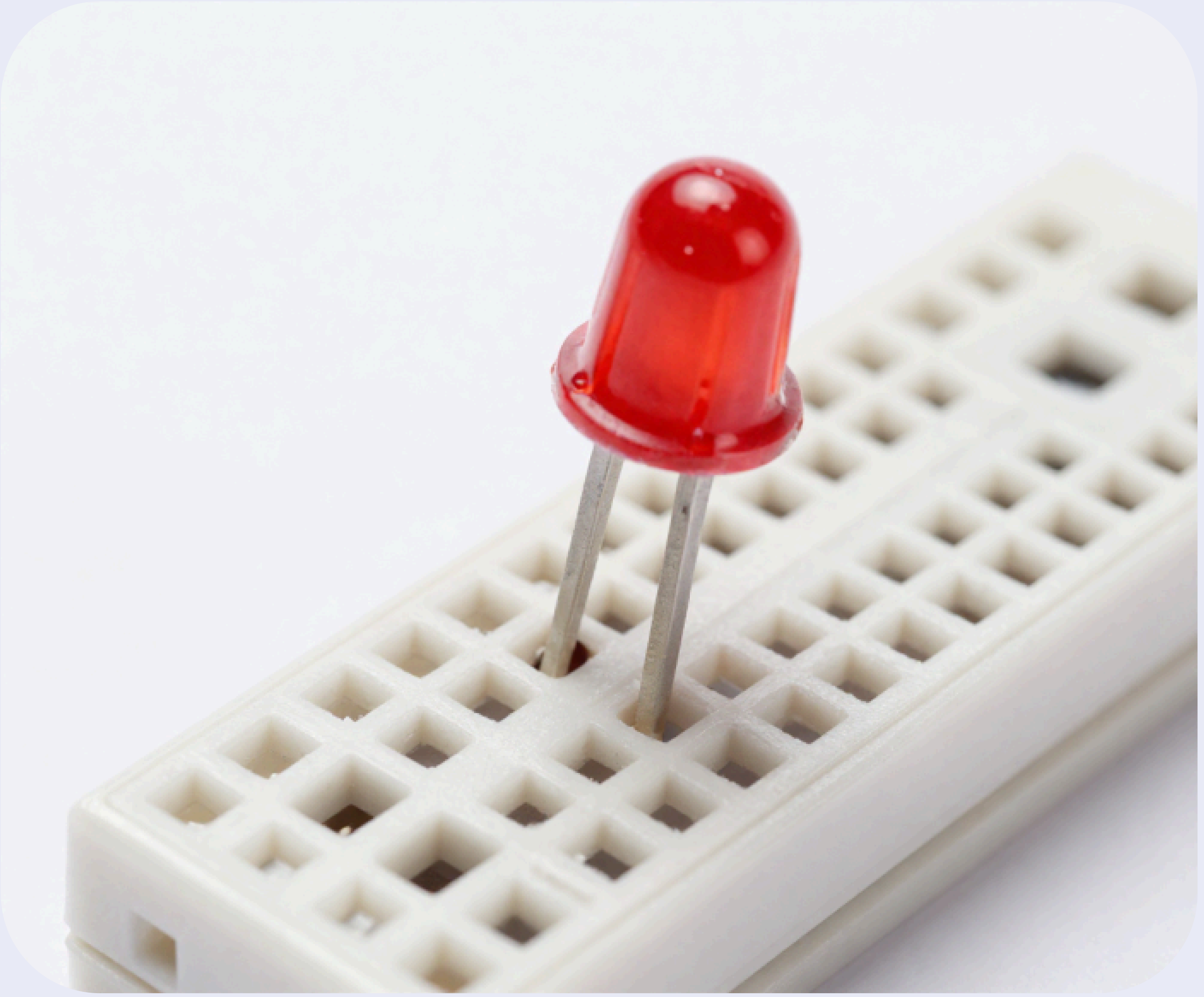


Step 2 — Insert the LED

Push the LED legs into the breadboard.

Remember: the long leg is positive (+) and the short leg is negative (–).

Make sure each leg goes into a different row!

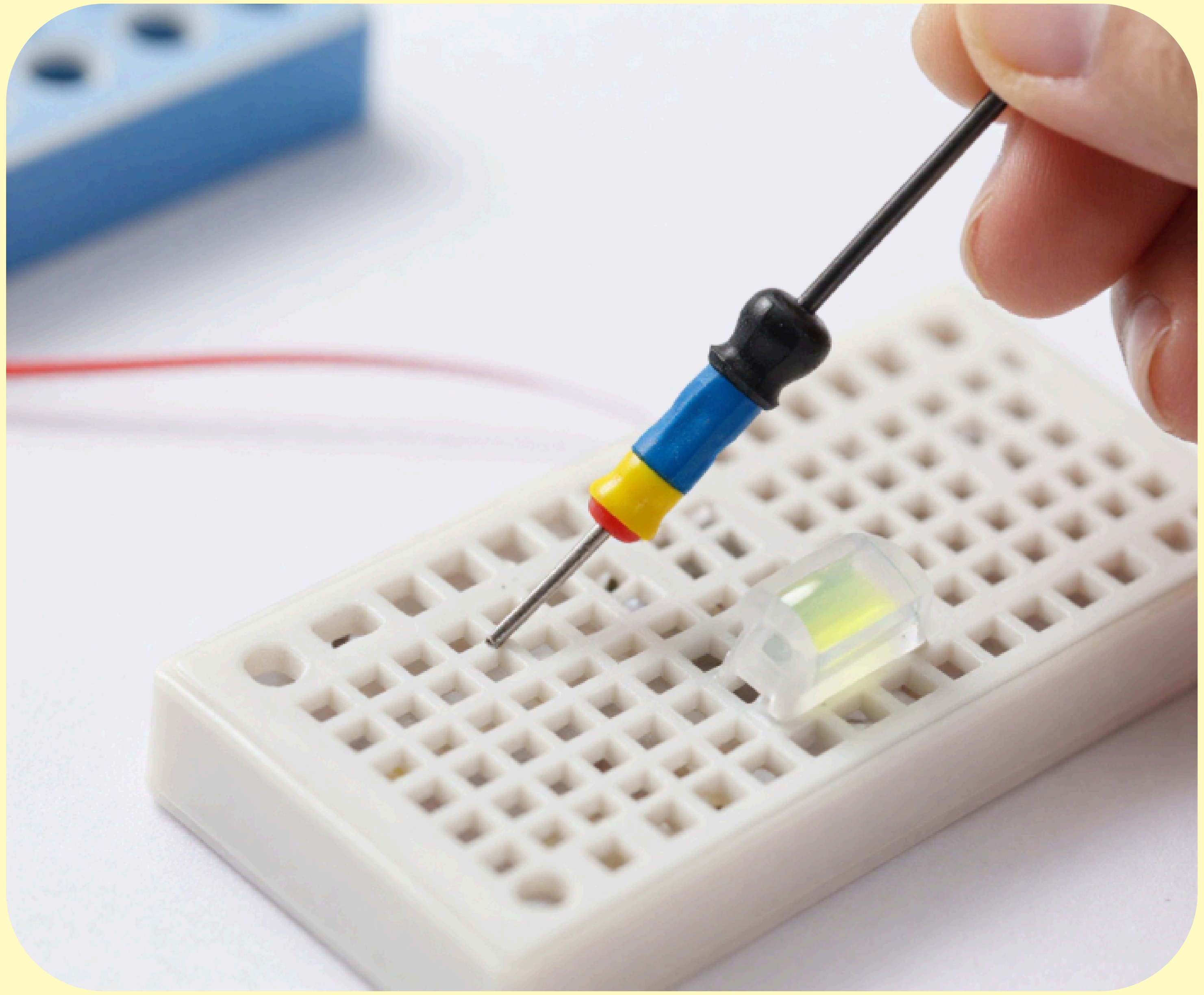


Step 3 — Add the Resistor

Connect the resistor from the LED's short leg row to the battery's negative side.

The resistor protects the LED from too much electricity.

Without it, the LED could burn out!

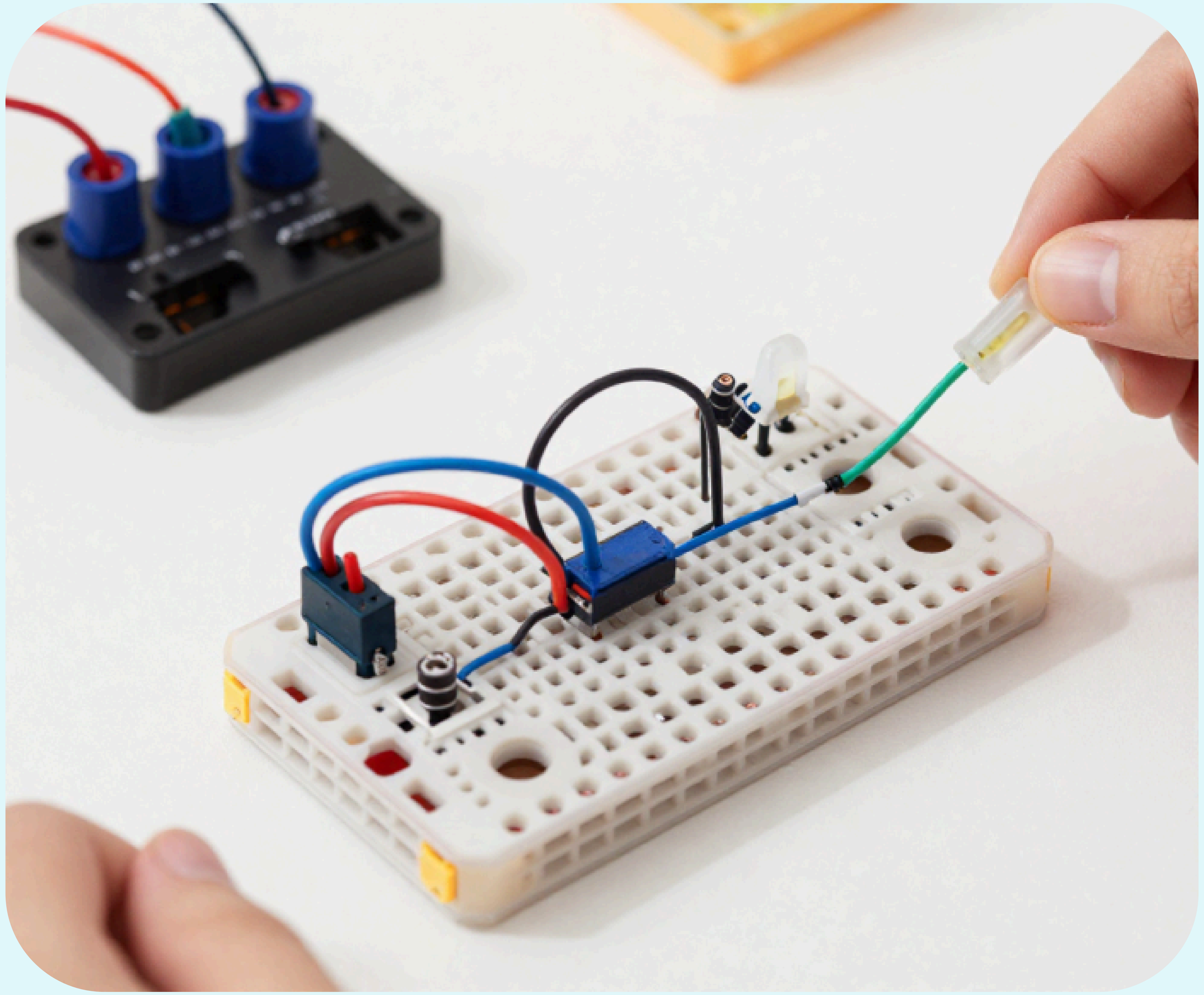


Step 4 — Connect the Wires

Use jumper wires to connect the battery to the LED and resistor on the breadboard.

Now electricity can flow and light the LED!

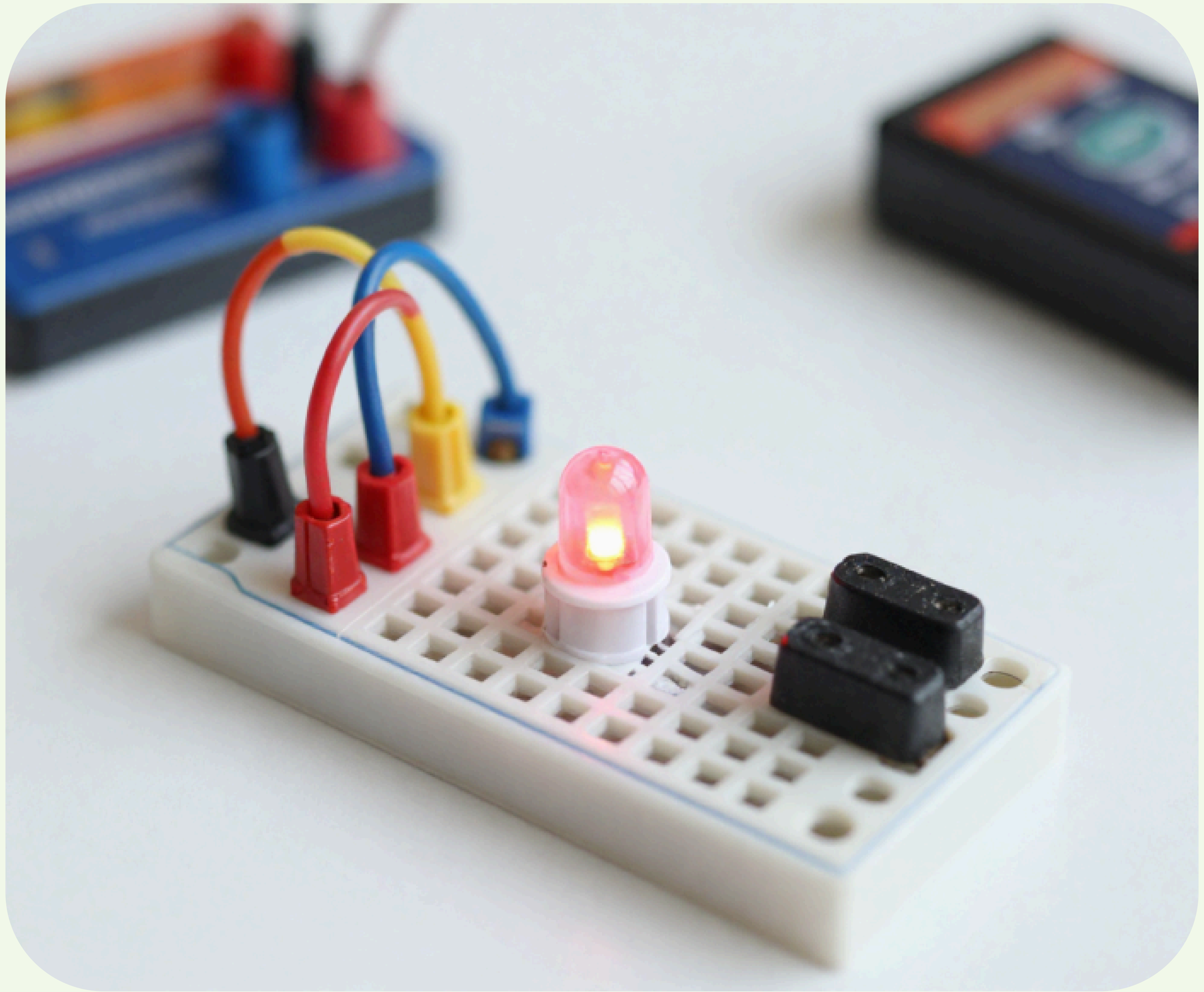
Check all connections before turning on.



Watch It Light Up!

Great job! Your LED lights up because electricity is flowing through your circuit.

The battery pushes electricity through the wires, resistor, and LED – making it glow!

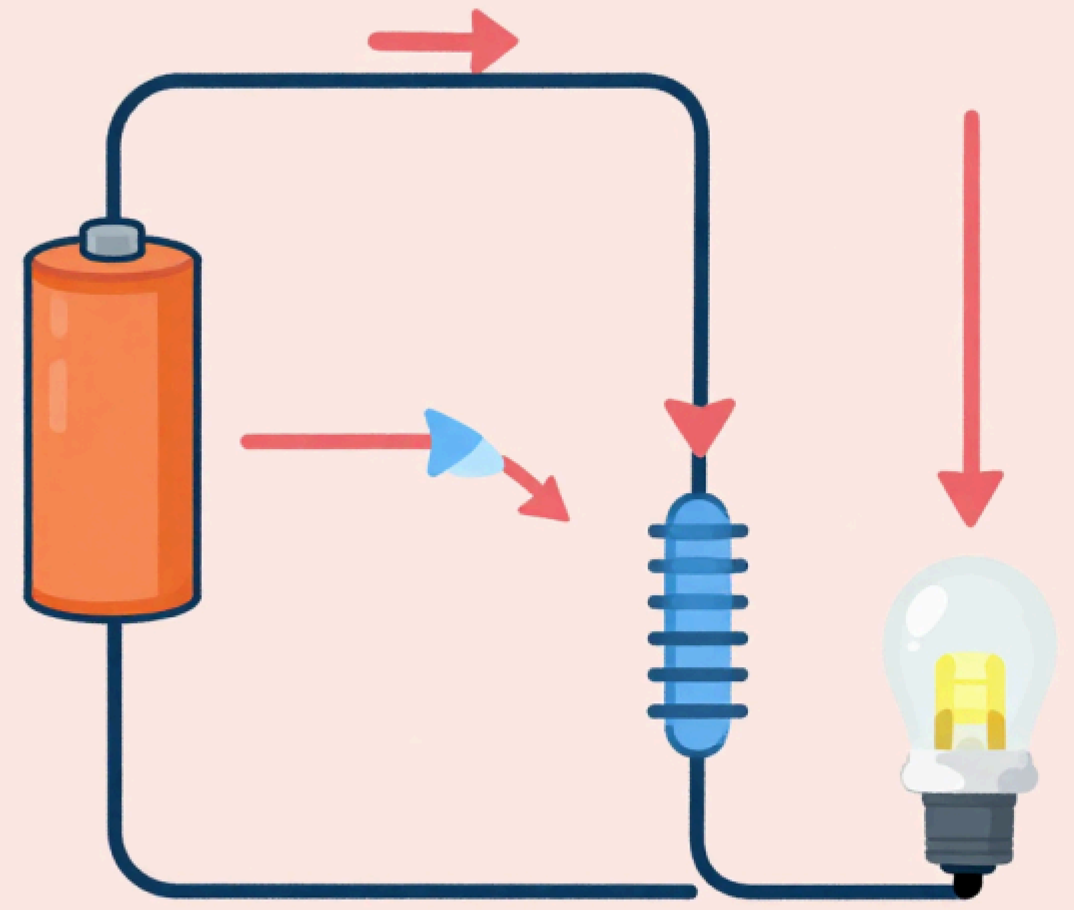


How Does It Work?

Electricity flows from the battery, through the wires, resistor, and LED, lighting it up!

The resistor slows down the electricity so the LED does not burn out.

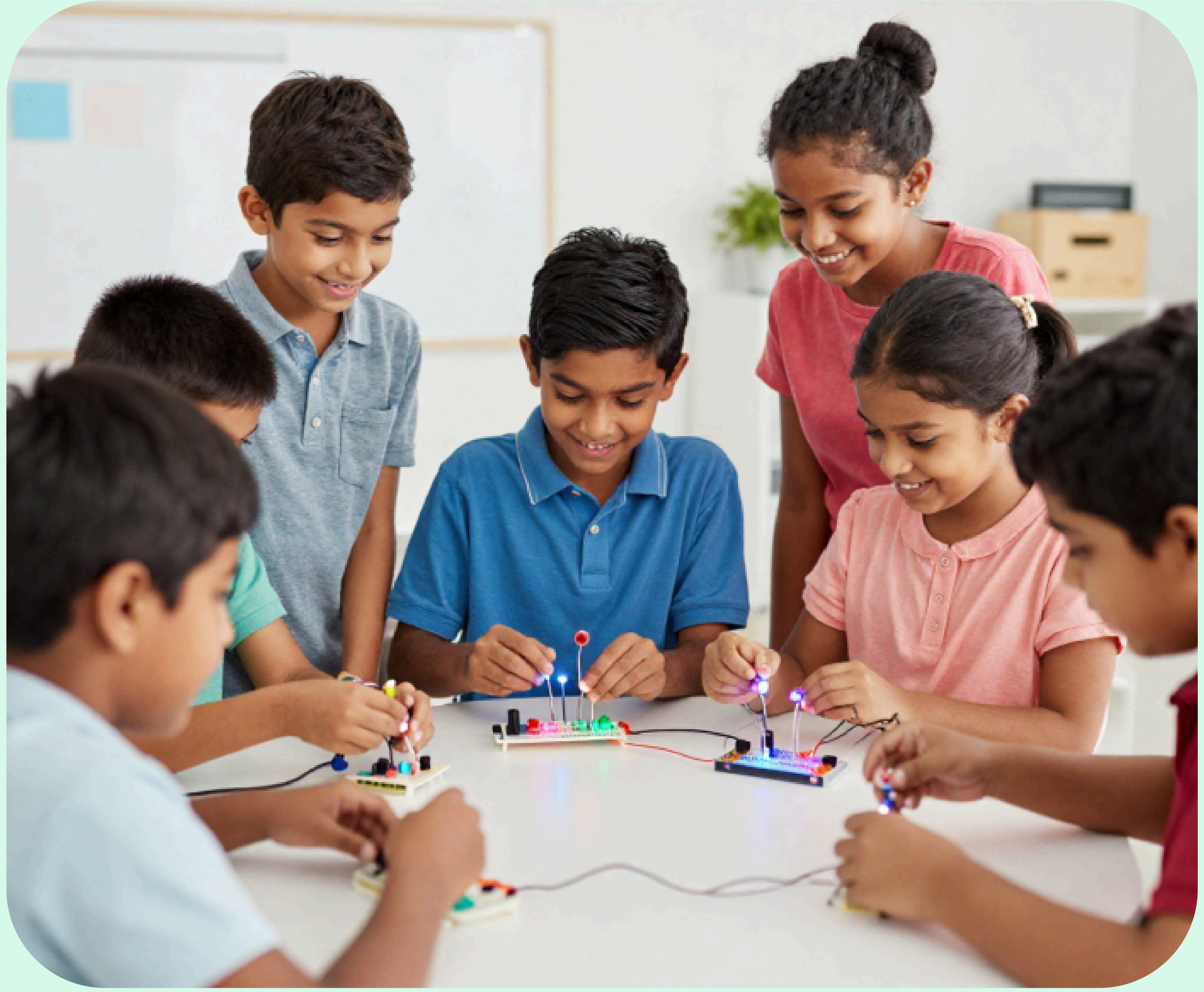
This path is called a circuit – a loop for electricity!



Try It Yourself!

Fun challenges to try:

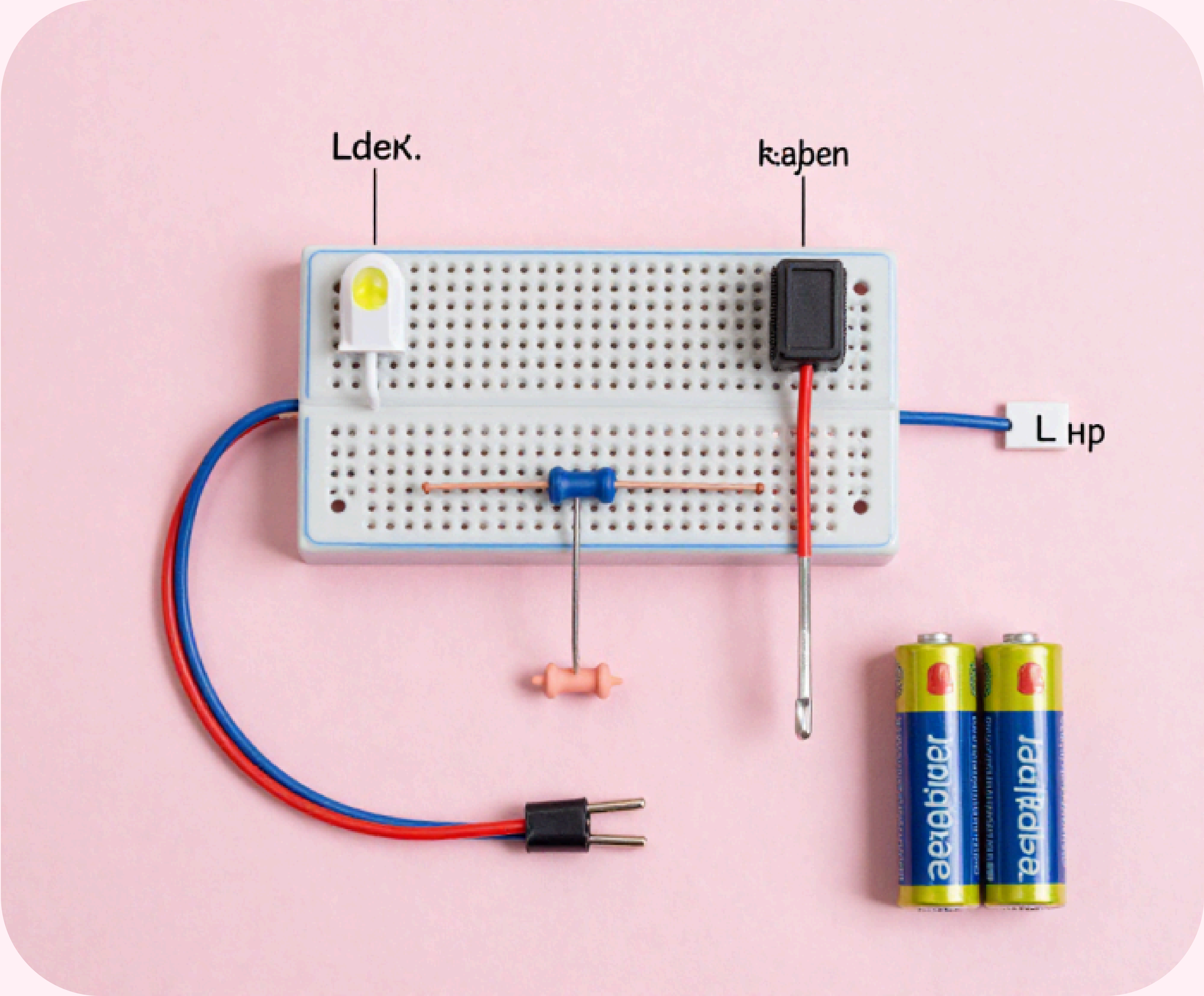
- Build the circuit again with a different colour LED
- Add a push-button switch to turn the LED on and off
- Try removing the resistor – what happens? (Ask an adult!)
- Connect two LEDs in a row



Recap: What We Learned

Key points:

- A breadboard helps connect parts without soldering
- LED lights up when connected correctly
- Resistors keep circuits safe
- Always follow safety rules
- Building circuits is fun and creative!



Keep Exploring Electronics!

Every inventor started with a simple idea – keep building and learning!

Next you can try:

- Making a traffic light with 3 LEDs
- Building a buzzer circuit
- Adding a motor to spin things

The world of electronics is waiting for you!

Thank You!

Ready to build your own circuits?

Ask your questions now!